Project | Sustainability Impact Analysis for Intel

INTRODUCTION: As you learned listening in on the strategy meeting with Dr. Alvarez and Intel's Sustainability Team, Intel is committed to reducing its carbon footprint and improving the sustainability of its devices – not just during manufacturing, but throughout the entire lifecycle.

A key part of this effort is their repurposing programs, which play a central role in achieving these sustainability goals. Repurposing and recycling programs aim to reduce e-waste, energy consumption, and CO_2 emissions by extending the life of existing devices, and thus reducing the need for new device manufacturing. Like Michael Campbell said: the average household in the US has anywhere from 3–5 PCs devices, tablets, notebooks, desktops that are perfectly functional, but not being used!

One challenge Intel faces is determining which devices in its repurposing program should be prioritized for the maximum environmental benefit. That's where data analysis comes in! To help with this, Intel gathered data on each device repurposed or recycled in 2024.

Your task is to evaluate the effectiveness of Intel's current repurposing strategy and provide a data-driven recommendation to help guide the program's direction and optimize sustainability efforts.

HOW IT WORKS: Follow the prompts in the questions below to investigate the data. Post your answers in the provided boxes: the **yellow boxes** for the queries you write and **blue boxes** for your text-based analysis. Once you're done, you'll submit your ***completed*** .pdf file to HQ for feedback from The Accelerator Team.

SQL App: Here's the link to our specialized SQL app, where you'll write your SQL queries and interact with the data.

NOTE: The dataset you are working with is designed for the Global Career Accelerator to reflect the key characteristics and structure of Intel's real data, while protecting their confidentiality and proprietary information. Be aware that any conclusions or results derived from this dataset should be viewed as hypothetical and for illustrative purposes only.

— Data Set **Descriptions**

In this project you'll query 2 different datasets, `intel.device_data` and `intel.impact_data`, that you will join together for your analysis. Here you'll find the data dictionary for each dataset.

`intel.device_data`

- `device_id`: Unique identifier for each repurposed device
- `device_type`: Type of device, values are either "Laptop" or "Desktop"
- `model_year`: The year the device was manufactured (e.g., 2018, 2019, etc.)

`intel.impact_data`

- `impact_id`: Unique identifier for the repurposed device's impact record (e.g., "LP20NA141592")
- `device_id`: Unique identifier linking the impact record to a specific device in the `intel.device_data` table
- `usage_purpose`: The specific purpose for which the device is being repurposed, values are Education & Digital Literacy, Corporate & Enterprise, Government & Public Sector, Environmental Sustainability Programs, and Social Impact & Non-Profit
- `power_consumption`: Power consumption of the device in watts (W) when in use (e.g., 50W, 75W)
- `energy_savings_yr`: Estimated energy savings per device per year when repurposed compared to a new device, measured in kilowatt-hours (kWh)
- `co2_saved_kg_yr`: Estimated CO2 emissions saved per device per year from manufacturing a new device, measured in kilograms (kg).
- `recycling_rate`: The percentage of the device that is recyclable (e.g., 80%, 90%).
- `region`: The geographical region where the device was repurposed, values are "North America", "Europe", and "Asia"

— **Task 1:** Organizing and Understanding the Data

We'll start by **joining** the device data with the impact data, allowing for a comprehensive analysis of device types, model years, repurpose regions, and energy savings in one dataset.

A. Simply write a query that returns all of the columns from both tables, joining the two on the `device_id` column. Be sure to choose the appropriate join so that all relevant

data is included in your result. **Note:** your query will have more than 150,000 rows (the max display for SQLPad!)

(paste your query below 👇)

```
SELECT *
FROM intel.device_data d
JOIN intel.impact_data i
  ON d.device_id = i.device_id;
```

B. To your joined dataset, add a new column called `device_age` calculated by subtracting the model_year from 2024. Paste your query below and double check that the values in your new column make sense. For example, a 2019 device should be 5 years old.

(paste your query below 👇)

```
SELECT *,
       2024 - d.model_year AS device_age
FROM intel.device_data d
JOIN intel.impact_data i
  ON d.device_id = i.device_id;
```

C. Order your joined data by model_year (oldest to newest). Do you notice more older (5+ years) or newer (under 5 years) devices being repurposed? What might that indicate?

(write your **answer** below 👇)

More newer devices are being repurposed: about 73% are under 5 years old, compared with 27% that are 5 years or older. The most common model year is 2021, suggesting devices enter the program on a typical 3–4 year refresh cycle.

D. Bucketing the `device_age` will allow us to analyze trends and patterns in energy savings and CO_2 reductions more effectively than using individual ages. Use a `CASE WHEN` clause to add one more column, called `device_age_bucket`, to your data, that is based on the device_age:

- `WHEN` the `device_age` is less than or equal to 3, `device_age_bucket` should be "newer"
- `WHEN` the `device_age` is greater than 3 but less than or equal to 6, `device_age_bucket` should be "mid-age"
- `WHEN` the `device_age` is greater than 6, `device_age_bucket` should be "older"

HINT: Instead of using e.g. `device_age <= 3`, you need to reference the calculation directly: `2024 - d.model_year <= 3`.

Double check that the values in your new column make sense! For example, a 2019 device should be characterized as "mid-age".

(paste your query below 👇)

```
SELECT d.device_type,
       d.model_year,
       i.*,
       2024 - d.model_year AS device_age,
       CASE
         WHEN 2024 - d.model_year <= 3 THEN 'newer'
         WHEN 2024 - d.model_year <= 6 THEN 'mid-age'
         ELSE 'older'
       END AS device_age_bucket
FROM intel.device_data d
JOIN intel.impact_data i
  ON d.device_id = i.device_id;
```

— **Task 2:** Key Insights

Now it's time to analyze the overall impact of Intel's repurposing program. You will use your final query from **Task 1** together with the `WITH` keyword for the remainder of this Project as you aggregate and analyze the data you've organized and prepped. For a refresher,

rewatch "🍿The `WITH` Keyword" in SkillBuilder 6.

A. What is the total number of devices Intel repurposed in 2024?

HINT: The dataset **is** representing all devices repurposed in 2024! You just need to COUNT all the rows in your joined data from Task 1!

(write your **answer** below 👇)

601,740 devices

B. Write a query that returns the total number of devices repurposed, the average age of repurposed devices in 2024, the average estimated energy savings (kWh) from repurposed devices per year, and the total CO_2 emissions saved (in tons) from repurposed devices.

Note: CO_2 emissions are typically measured in tons. Since `CO₂_saved_kg_yr` is measured in kg, divide the `SUM(CO₂_saved_kg_yr)` by 1000 to report the total CO_2 emissions saved in tons.

(paste your query below 👇)

```
WITH joined AS (
  SELECT d.device_type,
         d.model_year,
         i.*,
         2024 - d.model_year AS device_age,
         CASE
           WHEN 2024 - d.model_year <= 3 THEN 'newer'
           WHEN 2024 - d.model_year <= 6 THEN 'mid-age'
           ELSE 'older'
         END AS device_age_bucket
  FROM intel.device_data d
  JOIN intel.impact_data i
    ON d.device_id = i.device_id
)
SELECT COUNT(*)                    AS total_devices,
```

```
       AVG(device_age)              AS avg_device_age,
       AVG(energy_savings_yr)       AS
avg_energy_savings_kwh,
       SUM(co2_saved_kg_yr) / 1000  AS
total_co2_saved_tons
FROM joined;
```

C. Now that you have calculated the average estimated energy savings (kWh) and CO_2 emissions saved (tons), use ChatGPT to help put these numbers into perspective.

Try this prompt: I found that each repurposed device saves approximately of XXX kWh of energy per year and Intel's repurposing program saved XXX tons of CO_2 emissions in one year. Help me understand the significance of these numbers. How would this compare to the energy consumption of a small city or the amount of CO_2 produced by cars? What is the environmental impact of these savings?

What comparisons did you find most impactful in terms of scale? Summarize how much energy and CO_2 emissions were saved and how it compares to something familiar, like powering households or reducing car emissions.

(write your **answer** below 👇)

Each repurposed device saves about 25.7 kWh of energy per year, which is small on its own. Across all 601,740 devices, that adds up to a total of about 15.5 million kWh per year, more than enough to power roughly 1,475 US homes. The program also avoided about 6,768 tons of CO2, comparable to 1,470 cars off the road for a year. The most impactful comparison for me was the scale effect.

— **Task 3:** Identifying Trends & Maximizing Sustainability

By grouping our data in different ways, we can uncover patterns in energy savings and CO_2 reductions. These insights will help us determine which categories of devices contribute

the most to sustainability efforts and where Intel should focus its repurposing strategy for maximum impact.

A. Write a query that returns the total number of devices, the average energy savings, and the average CO_2 emissions saved (in tons), grouped by device_type.

Note (again): You'll need to divide `AVG(CO₂_saved_kg_yr)` by 1000 to report the average CO_2 emissions saved in tons.

(paste your query below 👇)

```
WITH joined AS (
  SELECT d.device_type,
         d.model_year,
         i.*,
         2024 - d.model_year AS device_age,
         CASE
           WHEN 2024 - d.model_year <= 3 THEN 'newer'
           WHEN 2024 - d.model_year <= 6 THEN 'mid-age'
           ELSE 'older'
         END AS device_age_bucket
  FROM intel.device_data d
  JOIN intel.impact_data i
    ON d.device_id = i.device_id
)
SELECT device_type,
       COUNT(*)                    AS total_devices,
       AVG(energy_savings_yr)      AS
avg_energy_savings_kwh,
       AVG(co2_saved_kg_yr) / 1000  AS
avg_co2_saved_tons
FROM joined
GROUP BY device_type;
```

B. Based on the results, which device type contributes the most to energy savings and CO_2 reduction? Why might that be the case?

Hint: Don't forget you can use ChatGPT as your Teammate to help think through your response!

(write your **answer** below 👇)

Laptops contribute the most energy savings and Co2 reductions. They make up about 68% of repurposed devices and therefore about 68% of the total savings. This is roughly 10.5 million kWh and 4,600 tons of CO2 per year. The two device types were nearly the same, so the difference comes from volume rather than efficiency. Laptops are likely more numerous because they are the most common device in workplaces and schools.

C. Write a query that returns the total number of devices, the average energy savings, and the average CO_2 emissions saved (in tons), now grouped by `device_age_bucket`.

(paste your query below 👇)

```
WITH joined AS (
  SELECT d.device_type,
         d.model_year,
         i.*,
         2024 - d.model_year AS device_age,
         CASE
           WHEN 2024 - d.model_year <= 3 THEN 'newer'
           WHEN 2024 - d.model_year <= 6 THEN 'mid-age'
           ELSE 'older'
         END AS device_age_bucket
  FROM intel.device_data d
  JOIN intel.impact_data i
    ON d.device_id = i.device_id
)
SELECT device_age_bucket,
       COUNT(*)                    AS total_devices,
```

```
       AVG(energy_savings_yr)       AS
avg_energy_savings_kwh,
       AVG(co2_saved_kg_yr) / 1000  AS
avg_co2_saved_tons
FROM joined
GROUP BY device_age_bucket;
```

D. Based on the result of your query, what do you notice about the relationship between device age and the number of devices repurposed versus the average energy saved?

(write your **answer** below 👇)

There is an inverse relationship between the number of devices repurposed and the energy saved per device. Newer devices are the largest group at about 53% but save the least per device, about 19.1 kWh per year. Older devices save the most per device but are only about 3% of the program. Mid-age devices fall in between at 32 kWh, and because there are 264,310 of them, they generate about 55% of the program's total energy savings. This suggests the prgram's current mix is weighted toward the devices with the lowest per-device benefit.

E. Finally, write a query that returns the total number of devices, the average energy savings, and the average CO_2 emissions saved (in tons), now grouped by region.

(paste your query below 👇)

```
WITH joined AS (
  SELECT d.device_type,
         d.model_year,
         i.*,
         2024 - d.model_year AS device_age,
         CASE
           WHEN 2024 - d.model_year <= 3 THEN 'newer'
           WHEN 2024 - d.model_year <= 6 THEN 'mid-age'
```

```
           ELSE 'older'
         END AS device_age_bucket
  FROM intel.device_data d
  JOIN intel.impact_data i
    ON d.device_id = i.device_id
)
SELECT region,
       COUNT(*)                      AS total_devices,
       AVG(energy_savings_yr)        AS
avg_energy_savings_kwh,
       AVG(co2_saved_kg_yr) / 1000   AS
avg_co2_saved_tons
FROM joined
GROUP BY region;
```

F. How does the carbon intensity of electricity in each region impact the total CO_2 savings from repurposed devices? Are there regions where repurposing leads to significantly higher environmental benefits? Why might that be?

(write your **answer** below 👇)

Average energy savings per device are almost the same in every region, about 25.7 kWh per year, but CO2 savings differ sharply. 15.5 kg per device in Asia, 10.3 kg in North America, and 6.4 kg in Europe. This is because of each region's electricity carbon intensity. Each kWh saved avoids about 0.6 kg of Co2 in Asia, .4 in North America, and .25 in Europe. Asia's grids rely more heavily on coal and other fossil fuels, so every kWh avoided prevents more emissions, while Europe's grid has more renewables and nuclear. As a result, repurposing in Asia delivers the highest environmental benefit. Asia has only 32% of the devices but 44% of the total CO2 savings, and one device repurposed there avoids about 2.4× the Co2 of one in Europe.

— **Task 4:** Data-Driven Recommendations

Using the findings from this analysis, we need to summarize key takeaways and develop actionable recommendations for Intel. Remember: the goal is to refine Intel's repurposing strategy to maximize energy savings and CO_2 reductions while ensuring the most effective use of resources.

A. Based on your analysis of the repurposed devices (including energy savings, CO_2 emissions, and device age), write **four** key takeaways in succinct sentences/bullets that summarize the most important patterns and insights from the data. These should be specific, concise, and focused on the implications of repurposing newer versus older devices.

(write your **answer** below 👇)

1. **Scale:** Intel repurposed 601,740 devices in 2024, saving about 15.5 million kWh of energy and 6,768 tons of CO_2 per year, even though each device saves only about 25.7 kWh on average.
2. **Older devices save far more per device:** Older devices (7+ years) save 48.0 kWh per year each, about 2.5 times the 19.1 kWh saved by newer devices (0–3 years), but they make up only 3.4% of the program.
3. **The program is weighted toward the lowest-impact devices:** Newer devices are 53% of everything repurposed but deliver only 39% of total savings, while mid-age devices (4–6 years) are 44% of devices and deliver 55% of savings.
4. **Region changes the CO_2 payoff, device type does not:** Laptops and desktops save nearly the same per device (25.8 vs. 25.6 kWh), but a device repurposed in Asia avoids 15.5 kg of CO_2 compared with 6.4 kg in Europe, because Asia's electricity is more carbon-intensive.

B. Based on your four key takeaways and ChatGPT as your teammate, write a recommendation for Intel on how to improve the repurposing program. Your recommendation should include a clear action or strategy for Intel based on the data and a data-driven justification for why this approach would maximize energy savings and CO_2 reductions.

(write your **answer** below 👇)

Intel should redirect its efforts toward reusing mid-age and older devices (4 years or older) and send more of them to areas with carbon-intensive electricity grids, starting with Asia.
Savings per device increase significantly as device age rises: 19.1 kWh for newer devices, 32.0 kWh for mid-age devices, and 48.0 kWh for older ones. Currently, over half of the program (53 percent) consists of newer devices, which provide the smallest amount of benefit. For every newer device replaced by a mid-age device in the program, annual savings rise by about 13 kWh (a 68 percent increase), and replacing a newer device with an older one yields about 29 kWh of savings. The region also affects the amount of CO_2 avoided: each kWh saved prevents about 0.60 kg of CO_2 in Asia, compared with 0.40 kg in North America and 0.25 kg in Europe. Yet Asia achieves 44 percent of the program's CO_2 savings even though it has only 32 percent of the devices.
Since there are few older devices (about 20,000 in 2024), it makes sense to focus on mid-age devices. Intel should focus its collection efforts on organizations with fleets of 4– to 6–year-old devices and expand its refurbishment partnerships in Asia. Device type is not a priority, as laptops and desktops yield almost the same amount per device.

C. Briefly reflect on how ChatGPT's suggestions influenced your recommendation. Did it help you see something you hadn't considered? What parts of your recommendation were improved based on its response?

(write your **answer** below 👇)

My expectations going in were that desktops would save more per device because they'd draw more power. The results showed the two types are nearly Identical, which moved my recommendation away from the device type.

— **LevelUp:** Optimizing Repurposing Strategy for Maximum Impact

Now that you've gained insights into the energy savings and CO_2 reductions across different device types and regions, let's use this data to optimize Intel's repurposing strategy for maximum environmental benefit.

A. Add to your final query of Task 3 that returns the total number of devices, the average energy savings, and the average CO_2 emissions saved (in tons), grouped by region, **the percentage** of the total energy savings and CO_2 reductions contributed by each device type within each region.

HINT: To calculate the percentage of the total energy savings, use this formula:
`Total energy savings for the device type / Total energy savings for the region) * 100`
You'll use a similar one for the percentage of the total CO_2 reductions.



Try this prompt: What's the best way to calculate the percentage of CO_2 reductions contributed by each device type in each region?

(paste your query below 👇)

```
WITH joined AS (
  SELECT d.device_type,
         d.model_year,
         i.*,
         2024 - d.model_year AS device_age,
         CASE
           WHEN 2024 - d.model_year <= 3 THEN 'newer'
           WHEN 2024 - d.model_year <= 6 THEN 'mid-age'
           ELSE 'older'
         END AS device_age_bucket
  FROM intel.device_data d
  JOIN intel.impact_data i
    ON d.device_id = i.device_id
```

```
)
SELECT region,
       device_type,
       COUNT(*)                    AS total_devices,
       AVG(energy_savings_yr)      AS
avg_energy_savings_kwh,
       AVG(co2_saved_kg_yr) / 1000  AS
avg_co2_saved_tons,
       SUM(energy_savings_yr) * 100.0
         / SUM(SUM(energy_savings_yr)) OVER (PARTITION
BY region) AS pct_energy_savings,
       SUM(co2_saved_kg_yr) * 100.0
         / SUM(SUM(co2_saved_kg_yr)) OVER (PARTITION BY
region)   AS pct_co2_savings
FROM joined
GROUP BY region, device_type
ORDER BY region, device_type;
```

B. Based on the results of your query, analyze the data to answer:

- Which device types in which regions contribute the most energy savings and CO_2 reductions relative to their numbers?
- How can this analysis help Intel prioritize specific device types in certain regions to maximize environmental benefits?

(write your **answer** below 👇)

Across device types, laptops account for slightly more than desktops in each region: although they make up about 67.7–67.9% of devices, they account for about 67.9–68.0% of the energy and CO_2 savings. The difference is small, so device type is not a major factor. The bigger variation is by region. Laptops in Asia achieve the highest CO_2 savings per device (15.5 kg per year), closely followed by desktops in Asia (15.4 kg), as compared to about 10.3 kg in North America and 6.4 kg in Europe. Although laptops in Asia represent only about 22% of all repurposed devices, they account for about 30% of total CO_2 savings.

Intel can prioritize by region first, then by device type. In Asia, increasing the repurposing of both laptops and desktops would achieve the greatest per-device reduction in CO_2, whereas in Europe the per-device return is lowest. Since the ratio of laptops to desktops is approximately 68/32 in each region and both types save almost the same amount of energy, Intel does not need to have a separate device-type strategy for each region; it simply has to take the type that is available and concentrate on the locations where the devices are being used.

C. In addition to focusing on sustainability, imagine Intel needs to optimize for cost-effectiveness in their repurposing program. How might you adjust your query to incorporate cost data (e.g., cost per repurposed device)? What strategies could Intel use to balance sustainability goals with cost constraints?

(write your **answer** below 👇)

I would add a third column with the cost of repurposing each device (for instance, the costs involved in collection, refurbishment, and shipping) and join it to my current data by device_id. I would also include cost-effectiveness indicators in the grouped query such as the average cost per device, the kWh of energy saved per dollar (calculated as SUM(energy_savings_yr) divided by SUM(repurpose_cost)), and the cost per ton of CO_2 avoided (calculated as SUM(repurpose_cost) divided by (SUM(co2_saved_kg_yr) divided by 1000)). When you group the results by region, device type, and age bucket, you will see which groups provide the greatest environmental benefit per dollar spent, rather than just the highest benefit per device.

— Evaluation Rubric

Unlike your Milestones that were evaluated largely based on your effort, the evaluation of your Portfolio Project will follow traditional evaluation methods, with tasks assessed for correctness and assigned point values accordingly.

Partial credit will be given where parts of this task are correct, even if other parts are incorrect or incomplete.

Task title	Max points
Task 1: Organizing and Understanding the Data	40
Task 2: Key Insights	25
Task 3: Identifying Trends & Maximizing Sustainability	60
Task 4: Data-Driven Recommendations	75
TOTAL POINTS:	200
LevelUp	
Optimizing Repurposing Strategy for Maximum Impact	20